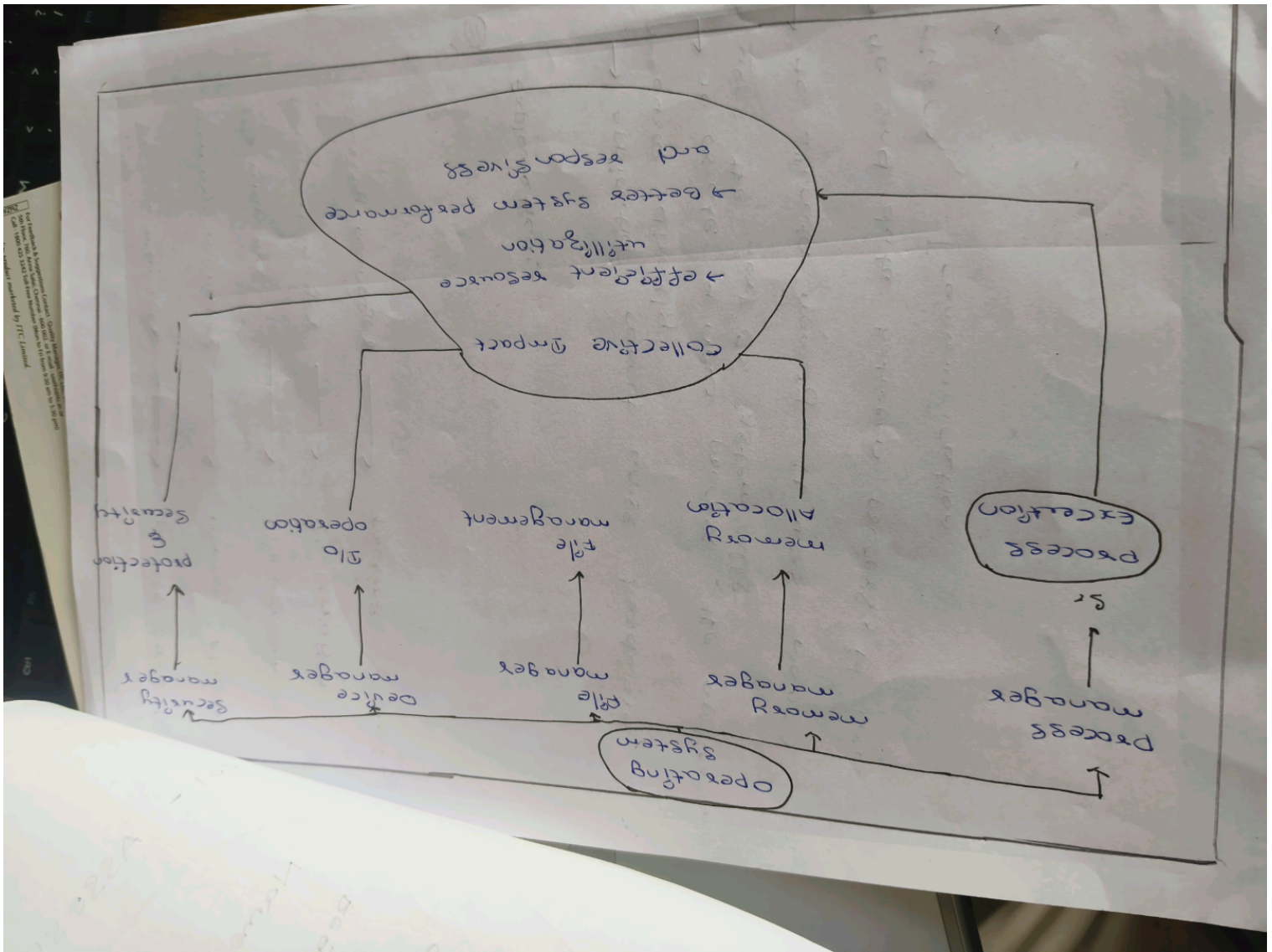




process manager	→	process execution
memory manager	→	memory Allocation
file manager	→	file management
device manager	→	I/O operation
security manager	→	protection & security

Explanation:-

- Application cannot access hardware resource directly
- To create a new process, the application uses the `fork()` system call
- To allocate memory, it uses `malloc()`
- To send data between process, it uses `pipe()` [communication]

Application program Requests services from OS through system calls

Create A new process

process control

Example System call

`Fork()`

Read a file

file management

Exam system call

`read()`

Allocate memory

memory management

example system call

`mmap()`

send data through a pipe

communication management

ex System call

`pipe()`

terminate exec

process management

ex System call

`exit()`

operating system kernel

operation to system calls mapping :-

operation	category	system calls
Create process	process control	fork()
Read file	file management	read()
Allocate memory	memory management	mmap()
Send data through pipe	communication	pipe()
exit process	process control	exit()